Hackathon Day 1 Progress Report

Project: AI-Based Academic Content Similarity Detector **Problem Statement:** Scan student research submissions against a sample academic-paper dataset using free, open-source similarity/embedding models to flag duplicates.

Submitted By:

Team Composition: 4 Members (2 from Computer Science, 2 from Mathematics)

1. Overall Team Progress Overview

Day 1 has been highly productive, and we have successfully moved from concept to a functional prototype. Our core achievements include:

- **Two Working Models Developed:** We have successfully built and initialized two distinct AI models utilizing open-source similarity and embedding techniques to scan and compare academic texts.
- **Prototype Website Live:** We have deployed a functional front-end prototype. The website serves as the user interface where research papers can be uploaded or pasted for similarity scanning against our sample dataset.
- **Core Architecture Established:** We have laid the groundwork for the data pipeline, ensuring that the text input from the website can be properly processed by our backend models.
- **Find Errors and Bugs:** During debugging we find some bugs and we are hoping to fix this before submission.

Team Leader Update:- (Krishna Sharma)

- **Prototype Live:** Developed and demostrated the initial prototype website for capturing student research submissions.
- **Model Generation:** Successfully built and initialized our fully functional open-source similarity and embedding models.
- **Workflow Established:** Aligned our interdisciplinary team, dividing core web infrastructure tasks to the CS members and high-dimensional vector/thresholding tasks to the Math members.

Team Member Update:- (Rishikesh Priyadarshi)

- **Prototype Live:** Help in developing the prototype website for research paper detection by doing backend work together.
- **Another prototype model**: I have also developed an 2nd prototype model for Research paper analysis in Window powershell.
- **Testing**: Help in testing the website and find error and bugs.

Team Member Update:- (asheesh yadav)

- **Data Collection:** They help in collecting data for analysis and debugging.
- **Research Paper Creation:** Both of them help us create multiple different type of research paper for analysis.

Team Member Update:- (Bishnu Tharu)

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